

A Discussion on Utilization of Heat Pipe and Vapor Chamber Technology as a Primary Device for Heat Extraction from Photon Absorber

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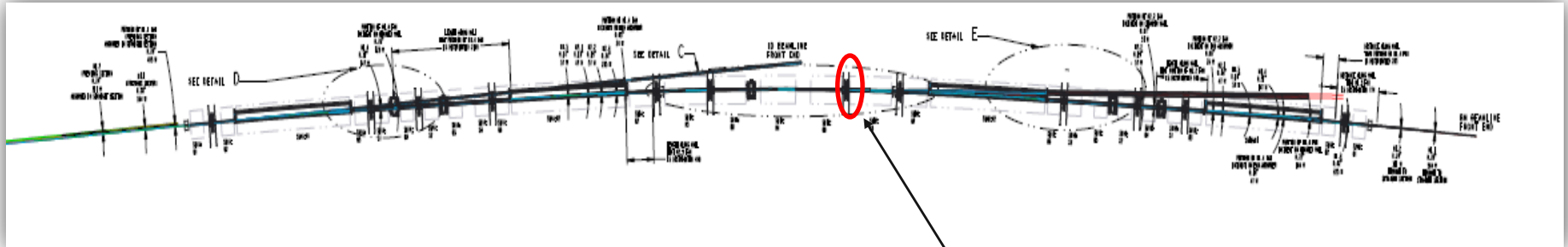


Outline

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- FEA (cont.)
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- Conclusions

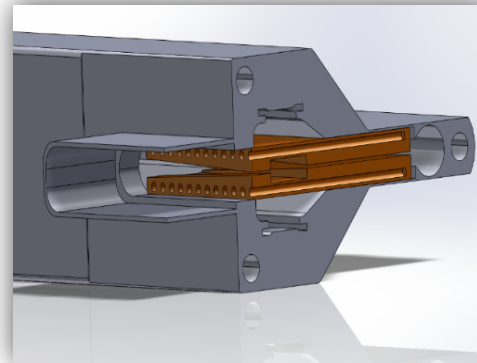


Problem Description

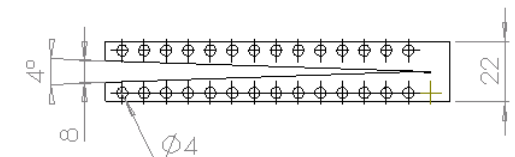
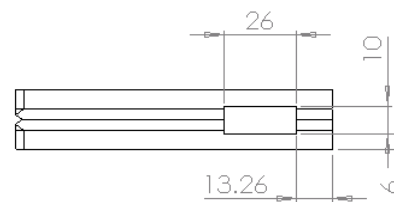
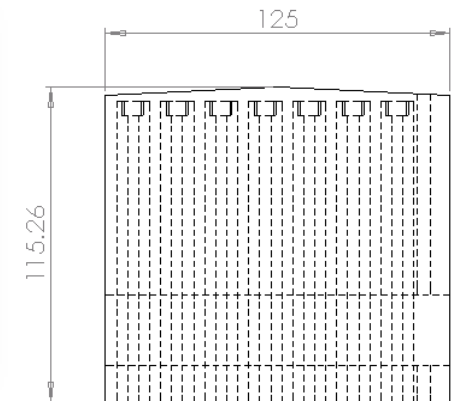


FEA Thermal Steady State Analysis [1]

- Current Design of B:CA1
 - 2.9KW
 - Limited space
- Problems
 - Max. Temperature
 - Temperature Gradient and Thermal Stresses



B:CA1 Absorber in Assembly [1]

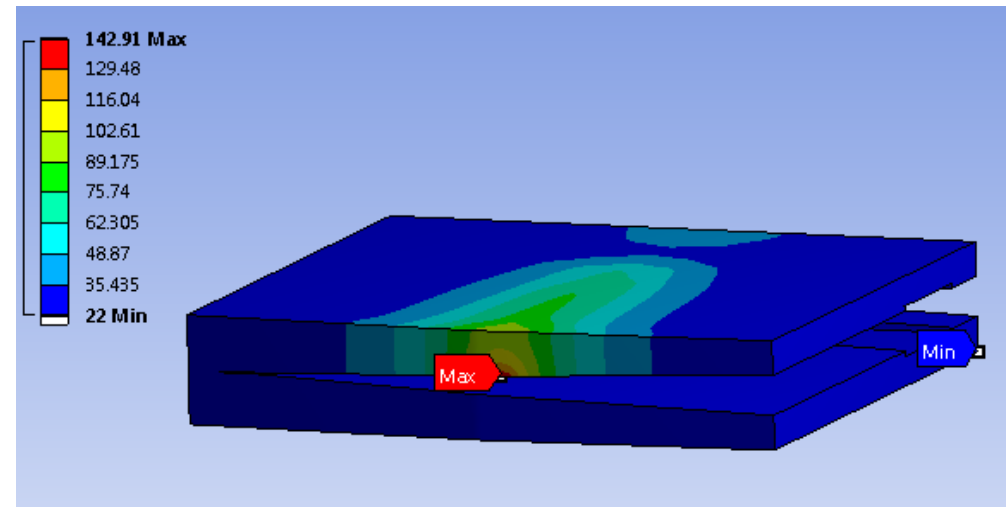


B:CA1 Drawing [1]

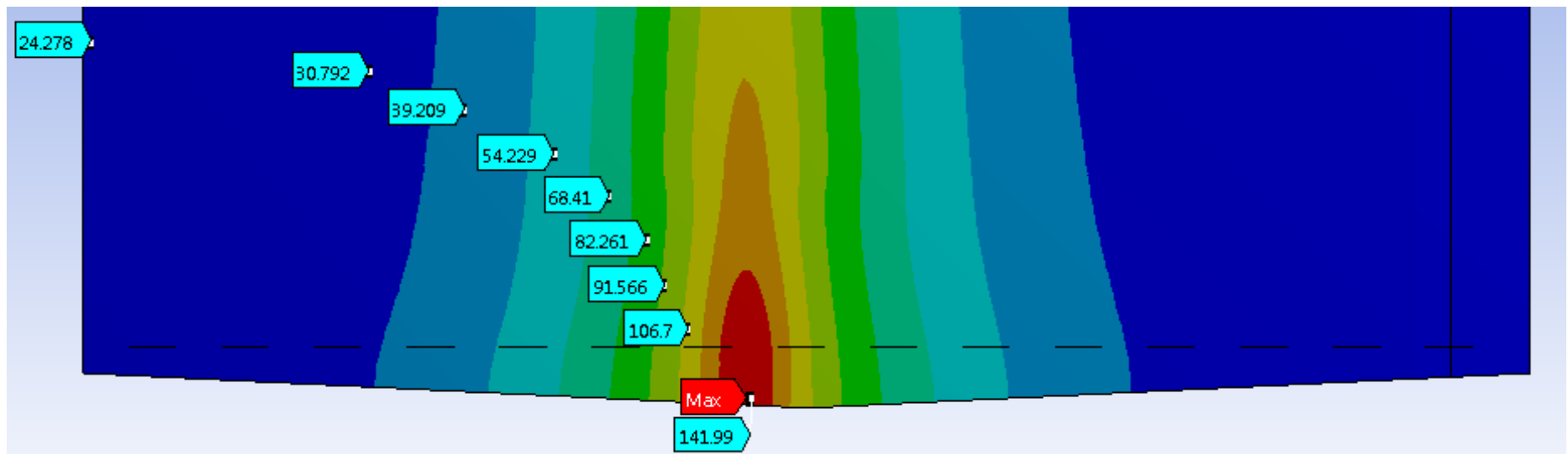
FEA Result of Baseline Design

■ Thermal Analysis of B:CA1

- Worst Case Scenario
- Risk Imposing Thermal Gradient
- High Maximum Temperature



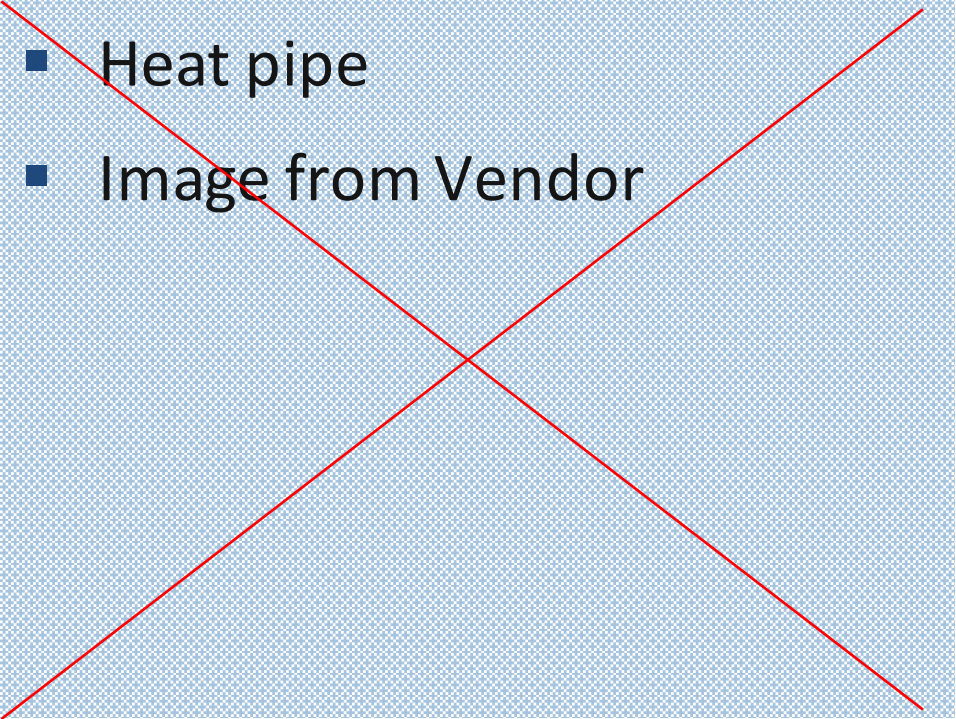
Water Cooled Thermal Steady State FEA



The Heat pipe



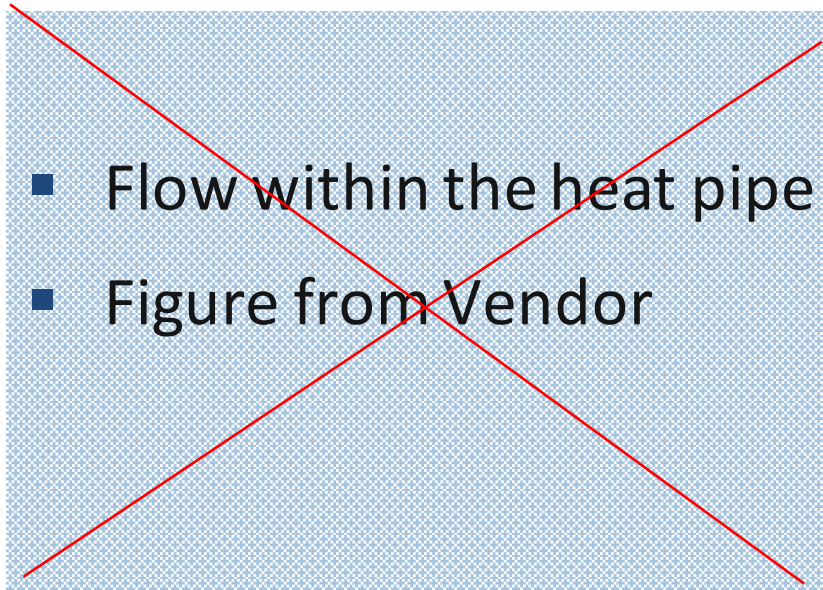
What is the Heat pipe?

- 
- Heat pipe
 - Image from Vendor

Ref: http://www.thermoguide.co.il/Heat_pipes.html

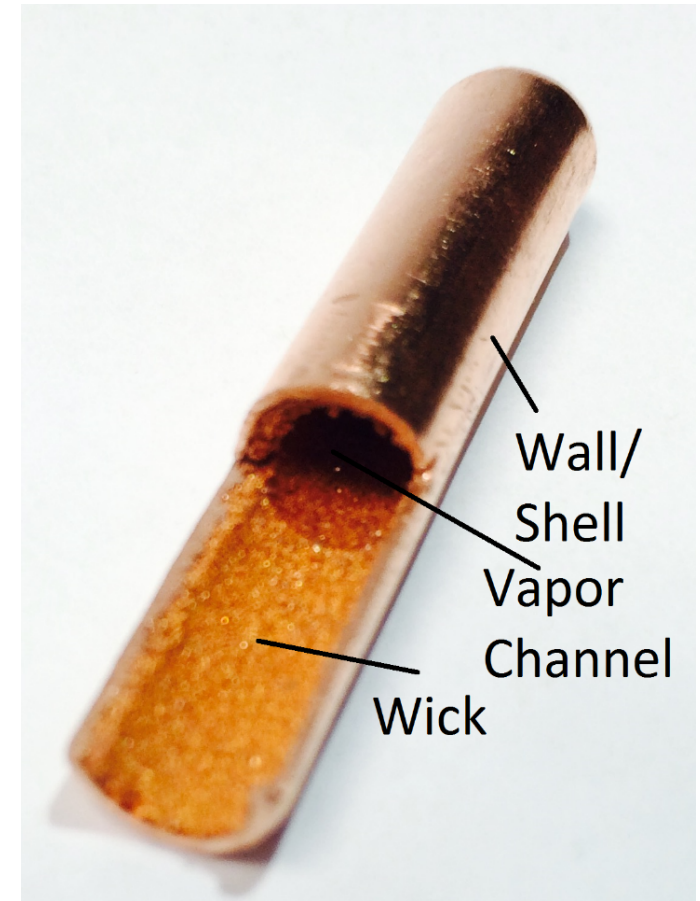


Heat Pipe Background



Heat Pipe Flow Diagram[2]

- Two Phase Heat Transport Device
 - Phase Change Phenomenon w/ Anti-Parallel Flow
 - Large Variety of Wick Types
 - Flexible Geometry
 - Low Vibration Expectations
 - **Equivalent** Thermal Conductivity that is 10 to 1000 times higher than metal in same shape



Copper-Water Sintered Powder Heat Pipe (Cut Open)

$$h_c = 5000 - 200,000 \text{ W/m}^2/\text{K}$$
$$k = 2000 - 100,000 \text{ W/m/K}$$



Advantage and Limitations



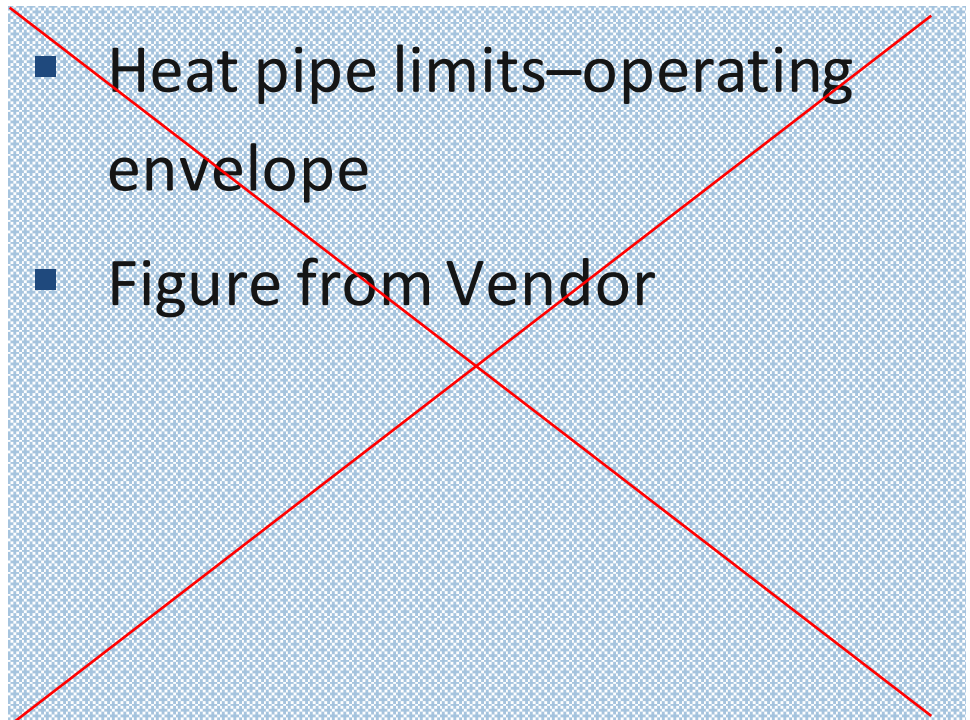
Typical Characteristics of heat pipes

Advantages:

- Very high effective thermal conductivity (**$\sim 100 \text{ kW/m/K}$**) compared to Cu (**0.4 kW/m/K**)
- **Vibration free** operation can be possible
- Can transport heat to a **large distance**
- They offer the possibility of **more compact design** and efficient means to remove heat
- Long Operating **Life (20+ years)** without any maintenance.
- Final heat removal can be done via conduction, or radiation heat transfer

Heat pipe limits-operating envelope

- Transport Limitations
 - Viscous/ vapor pressure
 - Sonic
 - Entrainment
 - Boiling
 - Capillary Pressure



Vapor Pressure or Viscous Limit

- Usually occurs at start-up
- The minimum pressure at the condenser end of the pipe can be very small due to cold or hot start up and difference in temperature.
- The **vapor pressure drop between the extreme** end of the evaporator and the extreme end of the condenser, represents a restriction in operation.



Sonic Limit

- At a temperature above the vapor pressure limit in the heat pipe, the **vapor** velocity can be comparable with **sonic velocity**
- At this point, the **vapor flow** within the heat pipe **becomes "choked"**.
- To avoid choked flow conditions (i.e., sonic limit) is to work below this limit of the maximum rate of heat transfer.

Reference: <http://www.thermopedia.com/content/835/>



Entrainment Limit

- The **vapor velocity increases** with temperature and may produce very high **shear force** on the returning liquid flow from the condenser to the evaporator, which cause entrainment of the liquid towards evaporator by the flow of vapor.
- The restraining force on **liquid** counter by **surface tension**, **that** is a major parameter in determining the entrainment limit. Entrainment will cause a starvation of fluid. Restriction of fluid flow from the condenser and eventual "*dry out*" condition at the evaporator.

Reference: <http://www.thermopedia.com/content/835/>



Circulation Limit

- The **driving pressure** for liquid circulation within the heat pipe is given by the capillary force established within the wick structure, given by following equation.

$$\Delta p_c \geq \Delta p_l + \Delta p_v + p_l g l \cos \phi$$

where Δp_l is the frictional pressure drop in liquid and Δp_v is the frictional pressure drop in the vapor.



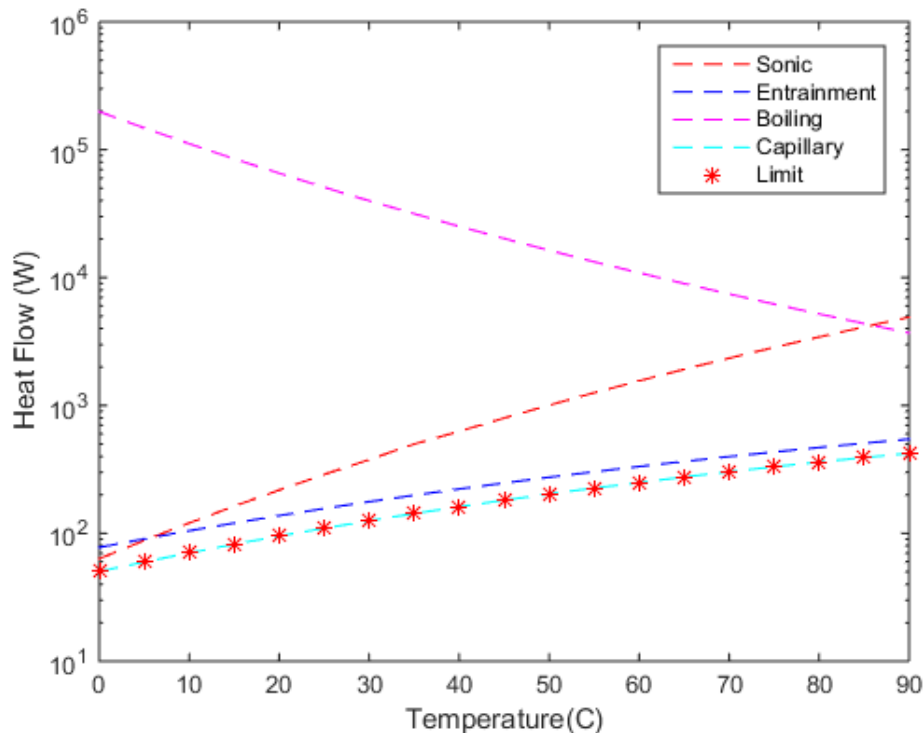
Boiling Limit

- Excessive heat input can create this condition. The **temperature drop** across the wick structure in the evaporator region **increases** with increase in incoming heat flux.
- A point is reached when temperature difference exceeds the degree of superheat that can not sustain in relation to nucleate boiling conditions. This condition of the **onset of Boiling** within the wick structure can **interferes with liquid circulation**. This can lead to the "*dry out*" condition, while in the case of constant heat flux, this can cause "**Burn Out**" of the evaporator containment.



Theoretical limits of Heat Pipe

Heat Transfer Limitations on Feasible Heat Pipe Design



Heat Pipe Limitation Plot

■ MATLAB

- 1000+ lines w/ GUI
- Outputs
 - Limitations Plot
 - Figure of Merit Plot
 - Maximum Heat Transport
 - Thermal Resistance
 - *Equivalent* Thermal Conductivity
 - Other characteristic values
- Optimization on Design Parameters

Finite Element Analysis

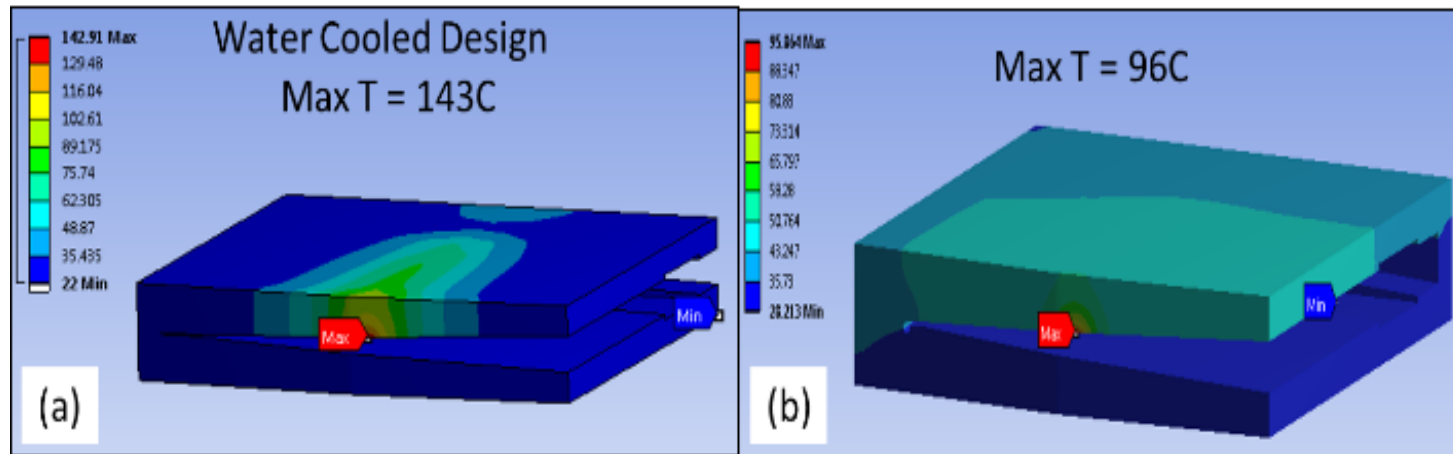


FEA cases

1. Pipe is modelled as a **hyper conductive rod** by equivalent thermal conductivity
2. **Multiple layers** of the heat pipe are modelled
3. Variations on the **hyper-conductive rod and multiple layer** model simulations are performed which treat temperature as a constant 2-5°C difference and alter the conductivities to achieve this.
4. Treat the **heat pipe's temperature difference** as a few degrees constant. Set the contact thermal conductance to change the thermal conductivity of the heat pipe to achieve desired temperature change.



FEA Comparison



- Generally, it was found that, regardless of the method used, a **heat pipe reduces the temperature** gradient on the absorber surface
- The **worst-cases scenarios** showed only a marginal improvement
- while others indicated a **larger benefit is achievable**, but all cases suggested that the heat pipe is a reasonable option in this application.

FEA Results

- **Maximum Temperature often lower**, but when higher it still is within a reasonable range and only in the overestimate methods
- **Temperature gradient** improved even in worst of models



Is it reliable?



RELIABILITY ISSUES

Start-up Conditions

- During start up the **temperature gradient** may vary considerably.
- Conditions such as – **non uniform distributions** of the fluid, which lack condensable gases resulting in a **sudden drop off in temperature**.
- **Failure is a legitimate concern** under such conditions.



Structural Stability

- It is important to note that heat pipes may be required to **withstand vibrations, shock, and severe temperatures.**
- Copper and GlidCop™, may be used to construct a heat pipe with very thin walls owing to the strengths of those materials.



Reliability Issues continued....

- Possibilities include: **capillary pumping failure, boiling, dry-out from a frozen fluids restricting the self-replenishing nature, and entrainment of the liquid flow.**
- In addition, **low thermal resistance at the condenser** can overwhelm the system and enhance the probability that such failures will occur.
- There are **ways to avoid some of these failure modes**, for example utilizing an entrainment limit which exceeds sonic limit.
- However, solutions to such problems, and the degree to which they may be avoided are strongly dependent on the **specific application.**



Manufacturing Concerns

- The development of a **shell**, the insertion of a **wick**, and filling the pipe with fluid as it is sealed off at one of the ends.
- Afterwards, some **experimentation** is needed on any given heat pipe to **ensure it performs as intended**.
- At the same time, to achieve the design needed, **the evaporative section needs to be built into the absorber** itself for all the heat pipes.
- The precise of manufacturing methods can also generate **heat pipe with different pressure** in each product and it does offer **concerns that arise from manufacturing process**.



Contact Variance

- Another major concern is the thermal interaction between the heat pipe and the absorber.



Material Compatibility for Operation

- Some fluid and solid combinations will result in chemical reactions that develop gas pockets which hamper or cease operation of the heat pipe.
- Water is known to be generally compatible with copper however compatibility between water and GlidCop™- another common absorber material, is less established.
- The Acetone or some refrigerants would be some potential alternatives by being compatible with both.
- However, these would offer notable downgrades in performance capabilities.



Solution?



Other Two-Phase Heat Exchanger Solutions

- Flat Heat Pipe (“Vapor Chamber”)

- Geometric Challenge
- Spreads Effectively

- Vapor Chamber image

Vapor Chamber [5]

- Pumped Two Phase Cooling

- Typically Isothermal
- High Heat Flux Capacity
- Requires Lower Flowrate

- Vapor Two phase cooling system image

Pumped Two Phase Cooling Diagram [6]



Conclusion

- **Heat pipes offer an attractive solution** for efficient and compact transport of heat in photon absorber applications.
- **Simulations** show that heat pipes offer **tangible improvements** over conventional water cooling.
- However, a number of **potential limitations** should be weighed against the performance improvements.
- These include potentially more complex and **costly fabrication, limited operational predictability, and reliability.**
- **Continued investigation is warranted**, particularly as the requirements for next generation accelerators continue to become more challenging.



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Questions?

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